

PANTHER® System

Distributor Informational Update- Top Failure Modes

Purpose

The purpose of this document is to provide service personnel with an Informational Update on the PANTHER Distributor. The number of customer complaints due to Distributor issues has spiked in recent months. This document outlines the problems that have been seen to date and their solutions.

Intended Use

This procedure is to be used by qualified personnel who may be required to service the PANTHER Systems.

What is Affected

Panther Distributor

What is Required

- Allen Wrench Set

Time Required

- Varies depending on inspection

References

- PANTHER Distributor Manufacturing Issues Informational Update
- GP11-228-FSN, PANTHER Incubator Pulley Replacement Belt Tensioning
- GP12-244-FSN, Replacement Procedure for Load Station and MagWash MTU Carriers.
- GP12-247-FSN Input Queue Screw Fastening and Inspection Procedure
- PANTHER System Service Manual
- PANTHER Troubleshooting Guide

Background Information:

The Distributor is currently at the top of the Panther failure mode pareto. This document will outline the top Distributor failure modes and how to remedy these problems.

Furthermore, remember that the Distributor has had numerous manufacturing issues. If the solutions outlined in this document do not resolve the issue, reference the *PANTHER Manufacturing Issues Informational Update* for a summary of Distributor manufacturing issues.

Also, due to these manufacturing issues, if a Distributor is ever replaced, give the spare part Distributor a full check to make sure there are no loose screws or other errors.

**Summary by Symptom and Location- organized by failure mode frequency.
(click to jump to the section)**

1. Distributor at Input Queue
2. Distributor Hook Step Losses at Modules with Carriers
3. Distributor Errors at Incubators
4. Distributor OBF Issues
5. Distributor Command Execution Timeouts
6. Distributor Thermal or Short Circuit Protection Activated or Distributor Voltage Out of Range error

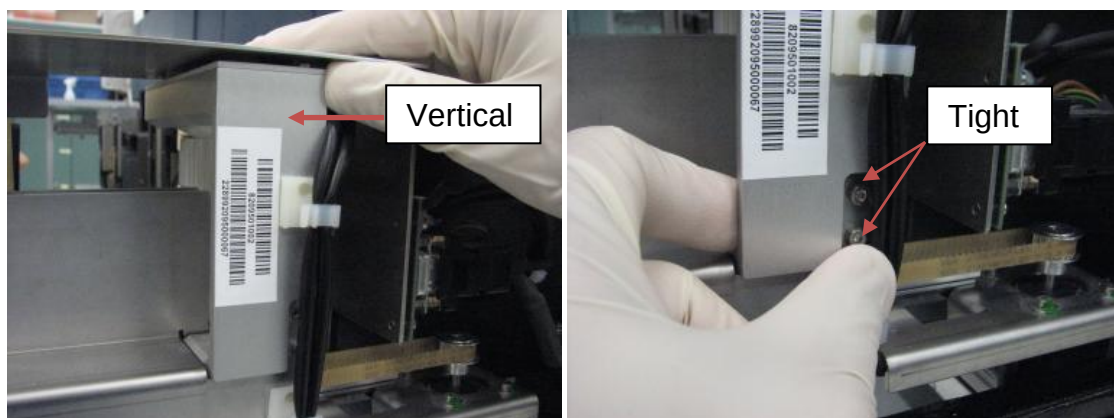
Remember- Distributor Step Loss errors come in 2 forms:

- **Distributor Lost Some Steps-** this means the Distributor lost some steps during the movement, but was able to recover. This error mode is indicative of a problematic location. When reviewing logs, if this error is ever seen, it should be addressed the next time the FSE attends the site.
- **Distributor Step Loss or Stalled-** This is a fatal error. The instrument was not able to recover from the step loss, and so the instrument crashed. It is helpful to observe the instrument in the crashed state to determine why the Distributor incurred the Step Loss, and if it was due to a jammed MTU, popped triplet, a movement error by the module the Distributor was interacting with, or another error.
- With any of the problems outlined in this document, Distributor teaching is always a possible root cause of the issue. If all other solutions have been exhausted, manually check the teaching of the Distributor to the offending module.

Failure Modes

1. Distributor at Input Queue

1. This is by far the Distributor's highest failure mode, with numerous repeat calls. Please be very careful when teaching the Distributor to the Input Queue.
2. The Distributor is taught to the Input Queue during manufacturing. Therefore, when installing the instrument, the Distributor teaching at the Input Queue should be checked, and **only changed if it is not correct**.
3. **Failure Modes:** The Distributor interaction with the Input Queue can have, in general, 3 possible failure modes:
 - a) Hook Step Loss (016.000.00128 or 016.000.00129)
 - b) Theta Step Loss (016.000.00096 or 016.000.00097)
 - c) After processing a pick MTU command, no MTU Present on Sledge (016.000.00055)
4. **Input Queue Stacking:** If any of these failure modes occur with the Distributor at the Input Queue, the first thing to check is if the Input Queue is stacking MTUs correctly.
 1. Visually inspect the Input Queue. Are the MTUs stacked correctly? Look for lifted up or tilted MTUs.
 2. Review the Event Log Entry for Input Queue stacking issues (080.000.00052).
 3. See the Panther Troubleshooting Guide for instructions on resolving Input Queue Stacking Issues.
5. **Input Queue Aluminum Head:** Check the Input Queue aluminum head (located close to the Luminometer). Confirm that the head is vertical and that the screws are tight:



6. **Input Queue- Loose Belt:** A loose Input Queue belt will lead to stacking issues and Distributor Step Losses. Confirm the Input Queue Belt is properly tensioned.
7. **Input Queue Drift:** If there are no Input Queue errors, then check to see if the Input Queue has drifted toward the back of the instrument. Repeated use of the Input Queue can cause the module to drift toward the Luminometer over time. When this occurs, the IQ and Distributor will no longer be aligned in the Y-Direction (Instrument front to back).
 1. Manually check the alignment of the Input Queue to the Distributor. (See the Panther Service Manual).
 2. The Distributor hook should sit between the two vertical lines drawn on the MTU.



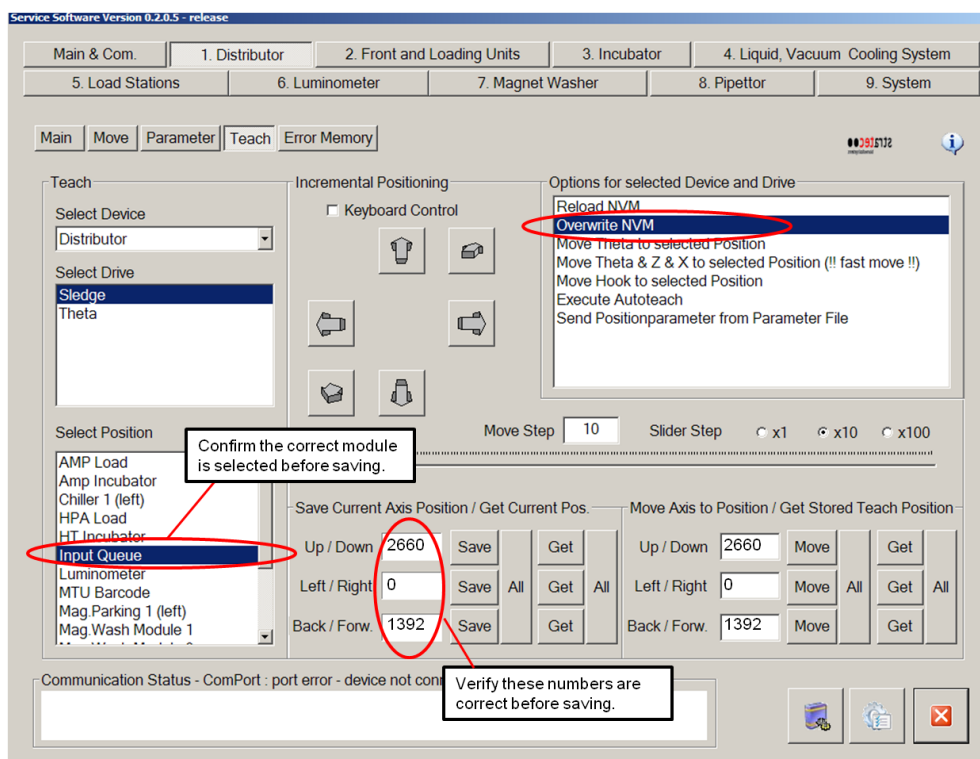
3. If the Distributor and IQ do not line up, adjust the Input Queue appropriately.
 4. Implement GP12-247-FSN to add lock washers to the Input Queue.
8. **Distributor Teaching** If there are no errors in steps 1-7, then the problem is probably a Distributor teaching issue. It is very important for the FSE to pay close attention to the teaching of the Distributor to the Input Queue to ensure that the teaching is accurate. (see the PANTHER Service Manual for a step-by-step guide on teaching the Distributor to the Input Queue)
 1. Historically, the 2 axes that have given the most difficulty teaching are the Hook Depth and the Z-Height.

1. Hook Depth: When extended, the hook should be **just barely touching** the MTU.
 2. It is important when teaching the hook depth that the FSE watches the hook extension. If the hook moves the MTU out of place at all, then the hook teaching should be adjusted (reduced). Then the teach MTU should be removed, and replaced on the IQ for another check. This should be repeated until the teaching is correct.
3. Z-Height: When extended, the hook should sit **just below the 7mm mark** that is drawn on the MTU.
2. When the teaching of each Distributor axis to the Input Queue is complete, the teach location needs to be saved. See Figure 1.

Reminders:

 1. When saving the teach location, the Distributor axes need to be in the desired teach location **before** saving.
 2. Confirm the Input Queue is highlighted in the *Select Position* list.
 3. Ensure the correct value is displayed in the position value box before selecting <Save>.
 4. Make the save permanent by double-clicking **<Overwrite NVM>**.
 5. Confirm the correct values were saved by moving the Distributor away, and then returning the Distributor to the Input Queue by selecting <Move Theta & Z & X to selected Position>, followed by <Move Hook to selected Position>. Confirm all axes are still aligned properly.

Figure 1- Saving the Distributor – IQ teach parameters



2. Distributor Hook Step Losses at Modules with Carriers

1. **Background-** For instruments 211 - 430, the MTU Carriers on the MagWashes, Sample Mix Station, Amp Load Station and HPA Load Station may be too tight. See Table 1 for when the Carriers were implemented by module:

Table 1

Module	Instrument #
Carrier - Sample Mix	258, 261+
Carrier - MW-Left	249+
Carrier - MW-Right	211+
Carrier - Amp Load	257+
Carrier - HPA Load	257+

2. The too tight MTU Carriers are especially harmful at the **Sample Mix Station** and the **MagWashes** because the tight MTU Carrier causes the tiplets in the MTU to pop up, causing a jam. See Figures 2 and 3. (Tiplets are no longer present at the Amp and HPA Load Stations).

Figure 2 Tiplet popped at carriage when pulling MTU out.

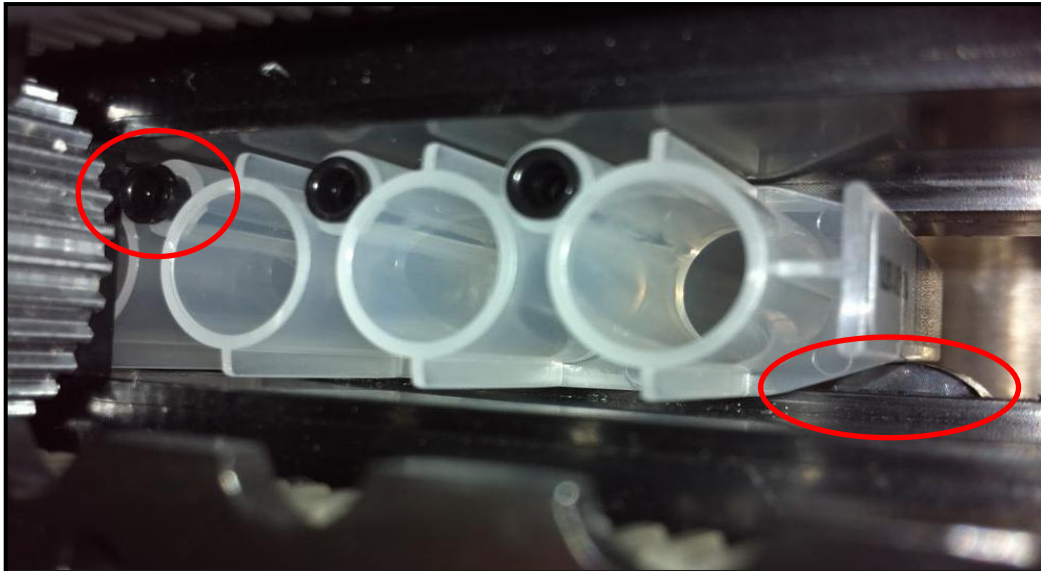


Figure 3- Triplet popped at carriage when placing an MTU



3. Solutions:

1. **Magwash:** Tiplet popups most often occur when the MTU is being removed from the module (Pick command). The only time that the MTU still has Tiplets when it is being picked from the MagWash is during Advance Clean. (During normal processing, the Tiplets are picked during the MagWash step.)
 1. Upgrade the instrument to software version 3.2.5 (or higher). This software changes the Advance Clean process to pick the tiplets immediately when the MTU is placed in the MagWashes.
 2. **Sample Mix Station:** If the tiplet popup failure occurs at the Sample Mix Station (or the Amp or HPA Load Stations or upgrading SW did not fix the issue at the MWs), it is best to replace the Module.

Note- The MTU Carrier will become a spare part as soon as October 15, 2012. When the MTU Carrier is released as a spare part, just the MTU Carrier should be replaced instead of the entire module. See GP12-244-FSN, Replacement Procedure for Load Station and MagWash MTU Carriers.

3. Distributor Errors at Incubators

Note that LD problems at the incubators can manifest with the following errors

- Hook Motor Step Losses
- Theta Motor Step Losses
- After processing a pick MTU command no MTU is present on sledge

1. LD at Incubators due to incubator step loss/ retry:

1. The majority of these errors occur at the Amp Incubator, especially in Software Versions SP2 and lower. In SP2 and lower, the Amp Incubator Motor current is set to a lower than optimal value.
2. Troubleshooting- First search in the logs for: “retries”, “lost”, “loss”. If those search terms do not return a hit, also search for:

Amp Inc: “COP->PC 22 52 01 02 09”

HT Inc: “COP->PC 20 52 01 02 09”

Trans Inc: “COP->PC 21 52 01 02 09”

This error should precede the Distributor Error. It is very important to look in the logs (log_PantherMain_YYYYMMDD...), and determine if there was an incubator error, as this failure mode can be difficult to diagnose.

3. Root Cause- Incubator positioning problems lead to a Distributor error. If the Incubator has a step loss, the Incubator will perform a retry. This retry causes the Incubator movement to be too slow, and the Distributor attempts to pick or place the MTU before the Incubator is properly positioned. When this occurs, there are a few Distributor failure modes:
 1. A standard Distributor step loss.
 2. If the Distributor is attempting to place an MTU, the MTU may get “double packed” i.e. the MTU is attempted to be placed in a slot that already has an MTU.
 3. If the Distributor is attempting to pick an MTU, the Distributor may miss the MTU completely.
 4. A “downstream error”. If there is a step loss at any incubator, it may throw the Distributor off, and the Distributor will attempt to place or pick in the wrong location, “downstream”, causing a fatal error.
4. Solutions-
 1. If the error occurred at the Amp Incubator, upgrade to Software 3.2.5 (SP5) or higher. This SW version increases the motor current to the Amp Incubator.

2. If the instrument is already at SW version 3.2.5, or the error occurred at the HT or Trans Incubators, then determine the reason for the Incubator retry. Improper incubator belt tensioning, poor pulley-belt meshing or a bad incubator pulley are common causes of incubator movement errors. See the *Panther Troubleshooting Guide* and FSN-GP11-228 for solutions.

2. LD Errors due to Incubator Slot Alignment

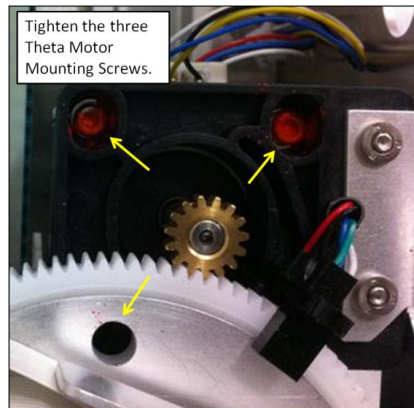
1. Troubleshooting- refer to the Panther Service Manual, *Incubator Slot Alignment* to determine if the Incubator Slots are correctly aligned to the door.
2. Root Cause- The alignment of the Incubator slots to the door is a taught parameter. If the alignment is not set correctly, the Distributor will have difficulty picking and placing MTUs at the Incubator.
3. Solution- Correctly align the Incubator slot to the door. See Panther Service Manual, *Incubator Slot Alignment*.

3. LD Errors due to warped Incubator slots.

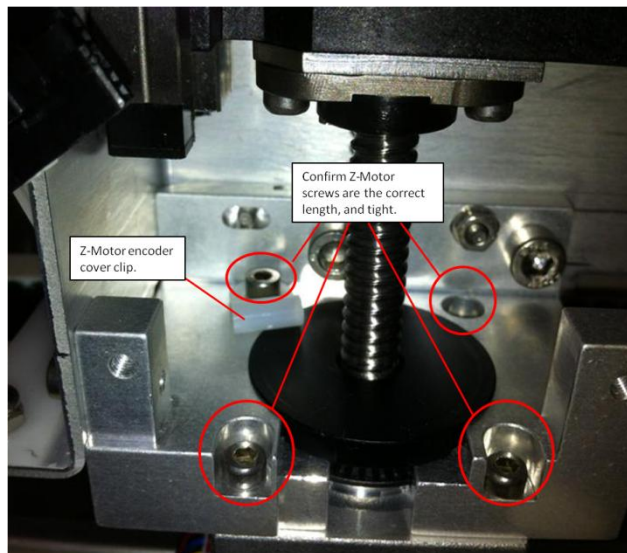
1. Troubleshooting- Find the slot the Distributor had difficulty interacting with. By hand, and using a MTU, feel if the slot is more rough than expected.
2. Root Cause- There have been some molding issues of the Incubator carousels.
3. Solution- Replace the Incubator.

4. Distributor OBF Issues

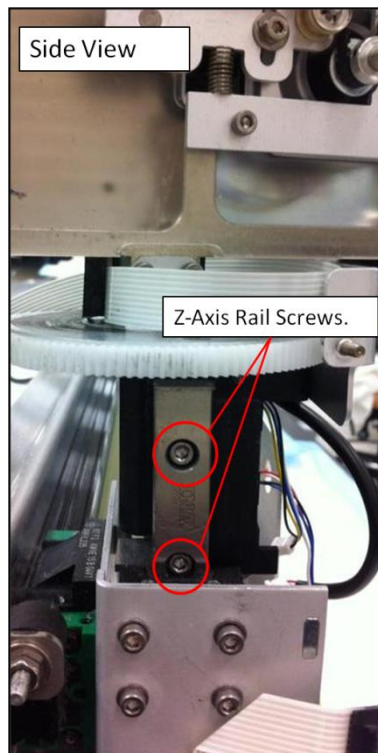
1. There have been numerous Distributor OBF issues. These problems and their solutions are fully described in the document, *PANTHER Manufacturing Issues Informational Update*. As a summary, be aware of the following Manufacturing issues:
2. **Loose Screws-** Numerous locations on the Distributor have suffered from loose screws. Listed here are the locations that have had documented loose screws, but all Distributor screws are suspect. Locations of note:
 1. Theta Motor Mounting Screws



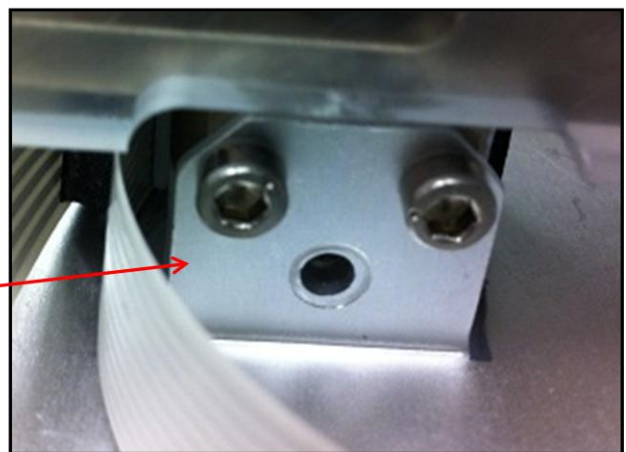
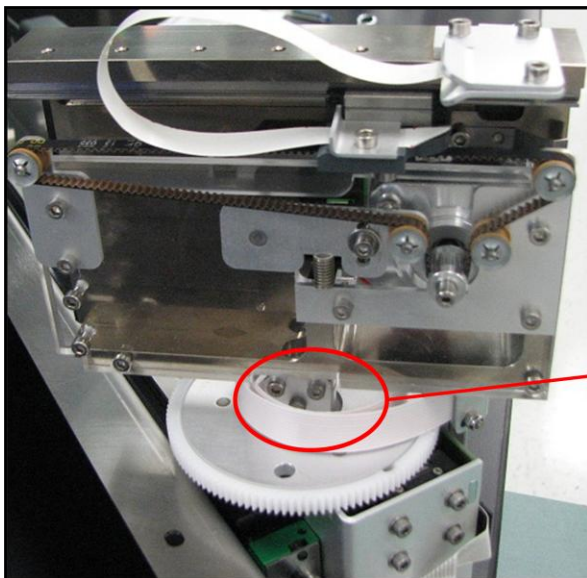
2. Z-Motor Mounting Screws (including the screw that holds down the Z-Motor encoder cover clip)



3. Z-Axis Rail Screws

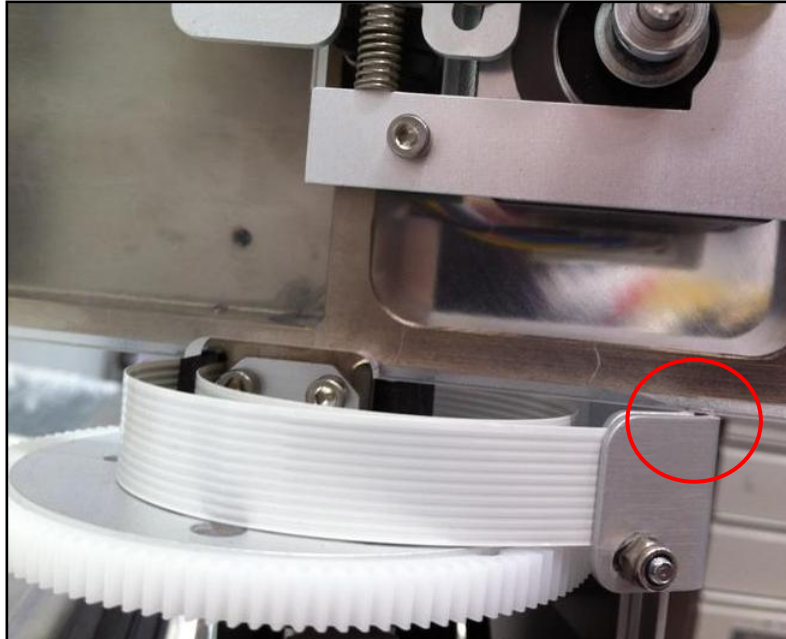


4. Head Bracket Screw(s)



3. Distributor Head Rubbing on Theta Cable Bracket

1. If the Distributor head is found to be rubbing on the Theta Cable Bracket, the bracket is most likely bent. Bend the bracket so the head no longer rubs along it.



4. Dirty Motor Encoders-

1. Dirty Motor Encoders will cause intermittent and unexplained motor step losses, especially during initialization. The two motors that have had the most problems are:
 1. Dirty Theta Motor Encoder
 2. Dirty Z Motor Encoder
2. See the *PANTHER Distributor Manufacturing Issues Informational Update* for information on solving these issues.

5. Broken/ dysfunctional Theta Motor Home Sensor

1. A broken or dysfunctional Theta Motor Home Sensor generally gives the symptom of the theta drive not being able to initialize. Often, the theta will rotate to the home position, but when the theta gets to the home position, the sensor will not recognize the home flag (hole in the disk), and so the theta drive continues to attempt to

initialize, past the home position, resulting in the motor stalling and a loud noise.

2. The Theta home sensor is not a spare part, so unfortunately, if the sensor is not working properly, then the Distributor module needs to be replaced.

5. Distributor Command Execution Timeouts

1. Command Execution Timeout errors due to the Distributor are rare, and can be difficult to repeat and diagnose.
2. Troubleshooting- In the logs, look for a 016.000.00014 message. This is a Distributor Command Execution Timeout error.
3. Root Cause-
 1. Confirm the Distributor is not jammed or stalled.
 2. Confirm all Distributor cables are plugged in properly.
 3. If no problems are found in steps a and b, then most likely the flex cable has gone bad. Since this problem is so difficult to reproduce, it is best to replace both the X-Axis and Theta-Axis flex cables. (See Panther Service Manual)

6. Distributor Thermal or Short Circuit Protection Activated or Distributor Voltage Out of Range error

1. Thermal or Short Circuit Protection Activated or Distributor Voltage Out of Range errors are rare. But, in general, they are both due to a bad PCB board, often due to the board getting wet.
2. Most frequently, the Distributor main PCB is the culprit, but the error should indicate which board.
 1. For example, "Distributor X drive thermal or short circuit protection activated" indicates the X board is bad.
 2. If more than one Distributor drive is giving the error, then the main PCB is probably the culprit.
3. When replacing the board, look to see if there are any obvious drips or splash marks. If so, locate where the drip/ leak came from on the instrument, and resolve that issue as well.